

Connecting to What You Know

In our last module, you worked with percentages — converting between decimals and percents, calculating percentage change, and finding amounts from percentages. Today's material builds directly on those skills. Discounts are simply percentages applied to prices, and once you can work with percentages confidently, discount math follows naturally.

You have probably encountered discounts as a consumer: "30% off" a jacket, or "buy within 10 days and save 2%." This pre-reading gives you the vocabulary and formulas to work with discounts the way businesses do — precisely and systematically.

Part 1: Trade Discounts

When a manufacturer or wholesaler publishes a **list price** (also called a catalog price), they rarely expect every buyer to pay that full amount. Businesses at different points in the supply chain receive different reductions from the list price, called **trade discounts**. A retailer might receive 30% off list. A large chain might receive 40%. The actual amount the buyer pays is called the **net price**.

The core relationship:

To find the net price, we use the **complement** of the discount rate. The complement is simply (1 minus the discount rate). If the trade discount is 30%, the complement is 0.70 — meaning the buyer pays 70% of the list price.

$$\text{Net Price} = \text{List Price} \times (1 - \text{Discount Rate})$$

Worked Example — Tidal Goods Co.:

Tidal Goods Co. orders wetsuits from a supplier. The list price is \$189.99 per unit, and Tidal Goods receives a 30% trade discount as a retail partner.

- Complement: $1 - 0.30 = 0.70$
- Net Price: $\$189.99 \times 0.70 = \mathbf{\$132.99}$
- Discount Amount: $\$189.99 - \$132.99 = \$57.00$